MIL-T-60506A(MU)
18 June 1970
SUPERSEDING
MIL-T-60506(MU)
5 August 1966

MILITARY SPECIFICATION

TELESCOPE ASSEMBLY: 10540506

1. SCOPE

1.1 This specification covers a telescope assembly which contains the transmitting and sight-receiving optics for the Range Finder, Laser, AN/GVS-3 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-C-22557/19	Connector, Receptacle, Electrical, Miniature (Screw on) Type UG-1619/U
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-810	Environmental Test Methods

FSC 1240

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

DRAWINGS

U.S. Army, Frankford Arsenal

F8565556	Pressure Tester - Two Station
F10540501	Range Finder, Laser, AN/GVS-3
F10540506	Telescope, Assembly
F10540514	Support and Caps Assembled and Machined
C10540586	Connector
F10540705	Housing
C10540723	Connector
C10540746	Connector
C10540748	Connector
B10540892	Connector
F10540967	Housing
F10544056	Fixture, Test Holding
F11727551	Support

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The telescope assembly shall be manufactured in accordance with Drawing F10540506 and drawings pertaining thereto.

3.3 General specification.- The telescope assembly shall meet the following requirements of MIL-F-13926:

a. Dimensions and tolerances
b. Inorganic protective surface finishes
c. Electrical and electronic assemblies
d. Soldering electrical connections
e. Part identification and marking
f. Workmanship

3.4 Environmental.- When the telescope assembly is subjected to the requirements of 3.4.6, the left and right sides shall be covered and sealed to simulate assembly as a component of the Range Finder, Laser, AN/GVS-3 during actual operating conditions.

3.4.1 Storage temperatures.- The telescope assembly shall be thermally stabilized at ambient temperatures of -80° Fahrenheit (F) and +160°F. Subsequently there shall be no evidence of damage to the telescope assembly, and it shall meet the requirements of 3.4.6 through 3.22.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

3.4.2 Extreme operating temperatures.- With the telescope assembly thermally stabilized at ambient temperatures of -65°F and +155°F, the requirements of 3.20 shall be met.

3.4.3 Normal operating temperature.- Unless otherwise specified, the telescope assembly shall meet the requirements of 3.4.4 to 3.22 inclusive while thermally stabilized at an ambient temperature between +70° and +80°F.

3.4.4 Vibration.- The telescope assembly shall withstand vibration for 15 minutes each in 3 mutually perpendicular directions. Vibration in each direction shall be at frequencies varied logarithmically from 5 to 500 cycles per second. The double amplitude of applied vibration shall be 0.036 inch at lower frequencies and shall be maintained at, but not exceed, a 5g level at higher frequencies. Subsequent to vibration, the telescope shall show no evidence of physical damage and shall meet the requirements of 3.4.6 through 3.22.

3.4.5 Shock.- The telescope assembly shall withstand a total of 18 shock impulses, three in each direction of 3 mutually perpendicular axes. Each shock impulse shall be a half-sine wave with a time duration of 11 ± 1 milliseconds. The peak amplitude of each shock impulse shall be 75g's. Subsequent to shock, the telescope shall show no evidence of physical damage and shall meet the requirements of 3.4.6 through 3.22.

3.4.6 Sealing.- The optical cavities of the transmitter and sighting telescopes shall be flushed with dry nitrogen until the gaseous mixture bled from each cavity has a dew point at least as low as -25°F. The pressure of the interior of each cavity shall then be raised to 5 pounds per square inch gage (psig) for a period of 5 minutes during which time the cavity shall show no evidence of leakage. The pressure within each cavity shall then be returned to normal atmospheric pressure and the cavity sealed. The interior atmosphere of each optical cavity of the accepted telescope assembly shall be nitrogen (see 3.4).

3.5 Electrical.- The electrical wiring of the telescope assembly shall conform to the wiring list contained on Drawing F10540506.

3.5.1 Continuity.- Each conductor of the telescope assembly shall be electrically continuous (nominally 0 resistance) between appropriate terminations as specified on the electrical schematic drawing.

3.5.2 Diode.- Forward bias diode characteristics shall be displayed when a positive voltage is applied to Pin L of Connector J2 and negative voltage is applied to Pin E of J3. Reverse diode characteristics shall be displayed when the polarity of the applied voltages are reversed.

3.5.3 Insulation resistance.- The insulation resistance between all mutually insulated points shall not be less than 1000 megohms when subjected to a minimum potential of 500 volts, direct current.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

3.6 Cleanliness.

3.6.1 Defects.- There shall be no evidence of moisture, grease, fingerprints, condensates, fractures and adhesive separations on any glass component when viewing through either end of the transmitter and sighting optical systems.

3.6.2 Dirt.

3.6.2.1 Sighting optical system.- There shall be no more than 2 particles of dirt visible in the central area of the reticle within the 5 mil mark, and the area of each dirt particle shall be not in excess of the area equivalent to a size 5 dig. There shall be no more than 5 particles of dirt visible in all the other areas of the reticle, and the area of each dirt particle shall be not in excess of the area equivalent to a size 10 dig. All observations shall be made from the eyepiece end. In addition, there shall be no more than 3 particles of dirt or foreign material (classed as dirt) on the surface of either objective or eyepiece lenses. The size of any dirt particle shall not exceed 0.05 inches.

3.6.2.2 Transmitter optical system.- There shall be no more than 1 particle of dirt visible on the surfaces of the negative and objective lens. The area of each dirt particle shall be not in excess of the area equivalent to a size 5 dig. All observations shall be made by direct viewing of the negative and objective lens respectively.

3.7 Functional operation with simulated range finder components.- The telescope assembly shall meet all functional requirements of this specification when operating with equipment simulating the components of the range finder.

3.8 Collimation.- With the face to which the laser cavity mounts, marked "Perpendicular to "A" within 0.0005" indicated on Drawing F10540511, Section C-C in a vertical plane, the requirements of 3.8.1 shall be met.

3.8.1 Transmitter optical system.- With an input beam perpendicular to the face, to which the laser cavity mounts, the beam shall emerge perpendicular to datum surface "A", indicated on Drawing F10540511, Section B-B within 1 mil in the horizontal meridian and within 2 mils in the vertical meridian.

3.9 Boresight adjustment.- The azimuth and elevation boresight adjustments shall each provide an excursion of the line of sight at least 3 mils to the left and 3 mils to the right, and 3 mils up and 3 mils down, from the position in which the line of sight is parallel (within 0.15 mils) to the line of sight of the transmitter optical system (see 3.8.1). The line of sight of the telescope through the center of the reticle pattern shall then be repositioned parallel in azimuth and elevation to the established line of sight of the transmitter optical system within 0.15 mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

3.10 Resolution.

3.10.1 Resolution (sighting system).- The resolution on the optical axis shall be six seconds of arc or less. Resolution consists of detectable line structure and proper line count in all four meridians. Focus shift between any two meridians shall not exceed 1/4 diopters measured at the telescope eyepiece.

3.11 Reticle accuracy.- The total angular subtense of the 80 mil azimuth span of the reticle as read in object space shall be 80 mils plus or minus 2.0 mils.

3.12 Eyepiece focus.- The diopter scale shall indicate zero diopters within plus or minus 1/4 diopter when eyepiece focus is adjusted to zero diopters. The range of focus from this position in each direction shall be at least 3 1/2 diopters as read on the eyepiece scale.

3.13 Parallax.

3.13.1 Parallax (sighting system).- Parallax in the center of the field of view of the sighting telescope shall not exceed 0.1 mil when viewing a target at a range of 1000 ± 100 meters.

3.13.2 Parallax (receiver system).- With the diaphragm (field stop) illuminated, parallax in red light shall not exceed 0.1 mil when viewed through a telescope, set for infinity, looking into the objective assembly. A Corning No. 70 filter or equivalent shall be used.

3.13.3 Parallax (transmitter system).- Parallax in red light shall not exceed 0.1 mil when viewing an infinity target positioned at the laser cavity mounting face. A telescope with a reticle adjusted for infinity focus in red light shall be used as the viewing device.

3.14 Focus (receiver system - field stop).- The field stop image shall be set for infinity focus within 1/8 diopter when viewed in red light with a dioptometer positioned behind the instrument laser filter 10540555.

3.15 Orientation.- The telescope assembly shall meet the requirements of 3.16 to 3.18 inclusive with the axis defined by the cylindrical mounting surfaces in a horizontal plane.

3.16 Plumb travel.- With the line of sight through the center of the boresight square of the reticle in coincidence with a plumb line at maximum elevation, the line of sight shall not deviate from the plumb line in excess of one mil when the telescope assembly is rotated to depress the line of sight 1000 mils. Total azimuth spread between any two elevations shall not exceed two mils.

3.17 Parallelism of reticle and image.- The vertical reticle line shall be parallel to the image of a plumb line within 2.0 degrees of arc when the line of sight is horizontal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

3.18 Image tilt.- The image of a vertical target line shall be vertical within 2 degrees of arc when the line of sight is horizontal.

3.19 Diaphragm.- The diaphragm aperture shall be in alignment with the geometric axis of the reticle (center square) within 0.1 mil and shall limit the field of view to the reciever to 0.75 mil ± 0.07 mil.

3.20 Operability.

3.20.1 Diopter adjustment.- The maximum torque required for rotation of the diopter adjustment for eyepiece focus shall not exceed 10 inch-ounces when the telescope assembly is thermally stabilized to the temperature requirements of 3.4.3 and shall not exceed 15 inch-ounces when thermally stabilized to the requirements of 3.4.2.

3.20.2 Potentiometer knob.- The maximum torque required for rotation of the potentiometer knob shall not exceed 5 inch-ounces when the telescope assembly is thermally stabilized to the temperature requirements of 3.4.3 and shall not exceed 10 inch-ounces when thermally stabilized to the temperature requirements of 3.4.2.

3.20.3 Boresight adjustment.- The maximum torque required for rotation of the azimuth and elevation boresight screws shall not exceed 3 inch-pounds when the telescope assembly is thermally stabilized to the temperature requirements of 3.4.3, and shall not exceed 5 inch-pounds when thermally stabilized to the temperature requirements of 3.4.2.

3.20.4 Parallax shield.- The parallax shield shall lock in detent in the following two positions: open horizontal position, and closed position. This requirement shall be met with the telescope assembly thermally stabilized to the temperature requirements of 3.4.2 and 3.4.3.

3.21 Reticle illumination.- With the reticle lamp energized with 28 volts, the reticle pattern shall be clearly defined when viewed in a darkened area. Rotation of the potentiometer knob clockwise from the extreme counterclockwise position shall complete the electrical circuit and then increase the intensity of reticle illumination from minimum to maximum.

3.22 Interface requirements.- The telescope assembly shall interface with:

a. Journal bearings of support and caps assembled and machined - 10540514 on yoke assembly.
b. Connector 10540586 (21P1) of yoke assembly
c. Mounting flange, locating pin hole, pilot flange and mounting holes of support 11727551 on cavity assembly.
d. Connector 10540892 (14J2) of cavity assembly.
e. Printed circuit board assembly (18J1) and mounting holes

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

f. Connector MIL-C-22557/19 (24J1) of voltage divider, and (15J1) of PFN power supply.

g. Mounting flange, locating pin holes, and mounting holes of Housing 10540967

h. Connector 10540746 (16P1) of nixie power supply (17P1) of PM power supply(22P1) of stop pre-amp; Connector 10540748 (15P1) of PFN power supply; and Connector 10540723 (25P1) of Counter (see Drawing F10540501).

i. Mounting flange, locating pin holes, and mounting holes of Housing 10540705.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 telescopes plus 3 each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPs, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. The tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS
(100% INSPECTION)

		REQUIREMENT	TEST PROCEDURE
101.	Electrical	3.5	4.6.3
102.	Continuity	3.5.1	4.6.3.1
103.	Diode	3.5.2	4.6.3.2
104.	Insulation resistance	3.5.3	4.6.3.3
105.	Cleanliness, defects	3.6.1	4.6.4.1
106.	Cleanliness, sighting optical system	3.6.2.1	4.6.4.2
107.	Cleanliness, transmitter optical system	3.6.2.2	4.6.4.3
108.	Resolution, sighting system	3.10.1	4.6.5
109.	Plumb travel	3.16	4.6.6
110.	Collimation, transmitter optical system	3.8.1	4.6.7.1
111.	Collimation, sighting optical system	3.9	4.6.7.2
112.	Boresight adjustment, azimuth	3.9	4.6.7.3
113.	Boresight adjustment, elevation	3.9	4.6.7.4
114.	Reticle accuracy	3.11	4.6.7.5
115.	Eyepiece focus	3.12	4.6.7.6
116.	Parallax, sighting system	3.13.1	4.6.7.7
117.	Parallax, receiver system	3.13.2	4.6.7.8
118.	Parallax, transmitter system	3.13.3	4.6.7.9
119.	Focus (receiver system - field stop)	3.14	4.6.7.10
120.	Parallelism of reticle and image	3.17	4.6.7.11
121.	Image tilt	3.18	4.6.7.12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

122.	Diaphragm	3.19	4.6.8
123.	Diopter adjustment, operability	3.20.1	4.6.9.1
124.	Potentiometer knob, operability	3.20.2	4.6.9.2
125.	Boresight adjustment, operability	3.20.3	4.6.9.3
126.	Parallax shield, operability	3.20.4	4.6.9.4
127.	Reticle illumination	3.21	4.6.10
128.	Sealing	3.4.6	4.6.2
129.	Interface requirements	3.22	4.6.11

MINOR: None defined

NOTE: The tests in Table I shall be conducted at a temperature between +70° and +80°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- Three telescope assemblies shall be selected at random as a special sampling from each 50 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Materials	3.1	Applicable dwgs & specs - visual
302.	Fabrication	3.2	Applicable dwgs - visual
303.	General specification	3.3	MIL-F-13926 - visual

NOTE: The tests in Table II shall be conducted at a temperature between +70° and +80°F.

4.4.2 Environmental.- Three telescope assemblies shall be selected at random as samples from each 50 produced or from each month's production, whichever occurs first. Each sample shall be subjected to the tests in Table III and shall then meet the requirements and tests in Table I.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Storage temperatures (-80° and +160°F)	3.4.1	4.6.1.1
305.	Extreme operating temperatures	3.4.2	4.6.1.2
306.	Shock (between +70' and +80°F)	3.4.5	4.6.1.3
307.	Vibration (between +70° and +80°F)	3.4.4	4.6.1.4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted items have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Performance of the telescope assembly.- Special testing equipment to test the telescope assembly, as shown in Figure 1, shall conform to the following requirements:

a. Leveling of special testing equipment.- The testing equipment shall be crossleveled utilizing a 10 second level positioned on Datum Plane "B" of the alignment adapter. Subsequent to the leveling procedure, Gage Block (Q) shall be positioned to Cavity Pin (P) to maintain the zero line of sight of the testing equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

b. Mirror adapter.- A front surface mirror shall be provided that will reflect the full aperture of the transmitter objective when oriented 45° to the line of sight of the transmitter optical system. The mirror face shall be set perpendicular to Datum Plane "B" of the alignment adapter within ten seconds of arc.

c. Holding device.- The Holding Device (W) shall position and clamp the telescope assembly as pre-set by the alignment adapter. The holding device shall provide an adjustable Gage Block (Q) to engage the Cavity Pin (P), "V" clamps, key slot for worm wheel, holding down screws and Leveling Screws (E). Design of the holding device shall assure repeatability of positioning the telescope assembly, subsequent to being pre-set by the alignment adapter.

d. Alignment adapter.- The alignment adapter shall be utilized to pre-set the Holding Device (W), collimators and gage block. The adapter shall be a representative of the critical elements and locating surfaces of the telescope assembly, i.e. journals, keyway arrangements, cavity mounting surface, Datum Plane "A" and the cavity locating pin. In addition, a Datum Plane "B" shall be provided as a horizontal reference plane. The adapter shall conform to the following accuracies:

(1) A plane passing through the axis of the pin and the axis of the journals shall be perpendicular to Datum "A" and the cavity mounting surface within 10.0 seconds.

(2) Datum "B" shall be perpendicular to Datum "A" and the cavity mounting surface within 10.0 seconds.

(3) The cavity mounting surface shall be perpendicular to Datum "A" within 10.0 seconds.

(4) Journals shall be concentric within .0005 TIR.

e. Collimators ("I", "J" and "K").- The Target Projector Collimators (J and K) shall have a clear aperture of at least 1.5 inches, and contain reticle test patterns capable of measuring the optical accuracies of the telescope assembly. The vertical reticle lines of the collimators shall be adjusted plumb. Collimator "I" shall be auto-collimated to Datum "C" and its line of sight aligned to the center of the transmitter optical system negative lens. Collimators "J" and "K" shall be auto-collimated to Datum "A". The line of sight of collimator "J" shall be aligned to the center of the objective lens of the sighting and receiving optical system. The line of sight of collimator "K" shall be aligned to the center of the transmitter optical system objective lens.

4.5.2.3.2 Sealing - telescope.- Special testing equipment to test sealing of the telescope shall conform to the internal pressure test requirements of Drawing F8565556, Pressure Tester - Two Station.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.5.2.3.3 Plumb travel.- Special testing equipment utilized for these tests shall conform to the requirements of Drawing F10544056.

4.6 Test methods and procedures.

4.6.1 Environmental.- The telescope assembly shall be thoroughly examined for physical defects after each environmental test. A visual and tactile examination shall be made for missing parts and physical damage. Upon completion of all environmental tests, the telescope assembly shall be subjected to the tests specified in 4.6.2 through 4.6.11 inclusive.

4.6.1.1 Storage temperatures.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 at the temperatures specified in 3.4.1. The telescope assembly shall be placed in a test chamber and the temperature of the test chamber reduced gradually (see note) to -80°F. The telescope shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually raised to +160°F. The telescope shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually reduced to room ambient temperature (+70° and +80°F). Caution shall be exercised during this test to avoid subjecting the telescope assembly to thermal shock.

NOTE: The rate of temperature change of the test chamber shall not exceed 20 degrees per hour throughout the temperature cycling tests of 4.6.1.1 and 4.6.1.2.

4.6.1.2 Extreme operating temperatures.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure I of MIL-F-13926 at the temperatures specified in 3.4.2. The telescope assembly shall be placed in a test chamber and the temperature of the chamber reduced gradually (see note) to -65°F. The telescope shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period and while at -65°F, the telescope assembly shall be subjected to the tests in 4.6.9.1 through 4.6.9.4 inclusive. Upon completion of this test, the chamber shall be gradually raised to +155°F. The telescope assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period and while at +155°F, the telescope assembly shall again be subjected to the tests in 4.6.9.1 through 4.6.9.4. Upon completion of this test, the chamber shall be gradually reduced to room ambient temperature (+70° to +80°F). Caution shall be exercised during this test ot avoid subjecting the telescope assembly to thermal shock.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.6.1.3 Shock.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 except the shock pulse shall comply with the wave form, time duration and peak "G" value specified in 3.4.5.

4.6.1.4 Vibration.- The test equipment utilized in this test shall conform to the vibration apparatus requirements specified in MIL-STD-810. The vibration test shall be performed in accordance with Test Method 514, Procedure I, Part I of MIL-STD-810, except the vibration equipment shall provide the conditions of amplitude, frequency, acceleration, time duration and directions specified in 3.4.4.

4.6.2 Sealing.- Internal pressurizing of the receiver and transmitter cavities to test for sealing shall be performed using dry nitrogen gas and special testing equipment conforming to 4.5.2.3.2. A dew point tester, with appropriate adapters shall be used to test the dew point of the interior atmosphere of each cavity. The cavities shall be purged by removing the purging screws and allowing the pressure used for testing to flow through each cavity of the telescope assembly until gas bled from the cavities conforms to the dew point requirements specified in 3.4.6. Subsequent to this test, each cavity shall be pressurized to the requirements of 3.4.6 in accordance with the procedure for internal pressure testing of Drawing F8565556. Retain the internal pressure in each cavity as specified in 3.4.6. At the conclusion of the specified time period, there shall be no evidence of leakage as specified in 3.4.6 as indicated on the pressure gage of the testing equipment. The internal pressure of each cavity shall then be reduced to normal atmospheric pressure and sealed as specified. Upon completion of this test, the left and right side of the telescope assembly shall be covered and sealed (see 3.4) to simulate assembly as a component of the range finder to test sealing of the telescope housing. The covers shall be designed to provide means for purging and pressurizing the telescope housing. Repeat the above test procedure to purge and test sealing of the telescope housing.

4.6.3 Electrical.- The wiring of the telescope shall conform to the schematic drawings and layout drawings utilizing standard measuring equipment and a visual inspection to determine conformance with the requirements of 3.5.

4.6.3.1 Electrical continuity.- A continuity test of the telescope shall be made utilizing standard measuring equipment to determine conformance to the electrical wiring and electrical schematic drawings specified in 3.5.

4.6.3.2 Diode.- The diode shall meet the requirements of 3.5.2 when checked with a standard volt-ohmeter.

4.6.3.3 Insulation resistance.- The telescope shall be subjected to the insulation resistance test in accordance with Method 302, Test Condition B of MIL-STD-202. Measurements between all mutually insulated points shall be made utilizing standard measuring equipment to determine conformance with the requirements of 3.5.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.6.4 Cleanliness.

4.6.4.1 Cleanliness - defects.- Perform this test in accordance with the requirements specified in 3.6.1. The accepted telescope assembly shall conform to the requirements of 3.6.1.

4.6.4.2 Cleanliness - sighting optical system.- Perform this test in accordance with the requirements specified in 3.6.2.1. The accepted telescope assembly shall conform to the requirements of 3.6.2.1.

4.6.4.3 Cleanliness - transmitter optical system.- Perform this test in accordance with the requirements specified in 3.6.2.2. The accepted telescope assembly shall conform to the requirements of 3.6.2.2.

4.6.5 Resolution (sighting system).- Position the telescope assembly on a "V" block capable of accepting the mounting bearings of the assembly. This test shall be performed utilizing an auxiliary telescope with a magnification of at least 3 power (3X), and a resolving power chart on a wall target. The resolving power chart shall represent the angular subtense for the seconds of arc specified in 3.10.1. The test method and procedure shall be in compliance with the "Resolution Test" specified in MIL-O-13830, and the resolving power chart shall contain four line sets as shown in Figure 1 therein. The auxiliary telescope shall be positioned at the eyepiece of the telescope assembly. Resolution of the sighting system shall meet the requirements of 3.10.1 as determined by viewing through the auxiliary telescope.

4.6.6 Plumb travel.- Perform the following test with the telescope assembly positioned on special testing equipment conforming to 4.5.2.3.3 and a calibrated wall target capable of measuring the excursion range accuracies for plumb travel of the telescope assembly as specified in 3.16. A gage block and the cavity locating pin of the telescope shall be utilized to level the telescope. Elevate the telescope assembly to +500 mils as indicated on the elevation scale. View through the eyepiece assembly of the telescope and establish coincidence of the actual intersecting point of a horizontal and vertical line (adjacent to the geometric axes) of the open boresight square of the telescope reticle with respect to the "0" vertical target image line and +500 mil line of sight. While observing the established position of the reticle with respect to the "0" vertical target image line, rotate the elevation knob of the testing equipment to depress the line of sight to -500 mils as indicated on the elevation scale of the telescope. During this rotation, the intersecting point of the reticle lines shall track the "0" vertical target image line within the tolerance specified in 3.16. The total deviation between any two error readings, throughout 1000 mils of excursion, shall not exceed the value specified in 3.16 as measured on the target image.

4.6.7 Performance of telescope assembly.- Perform the following tests with the telescope positioned on special testing equipment conforming to 4.5.2.3.1 and Figure 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.6.7.1 Collimation - transmitter optical system.- Observe through the viewing Collimator I and transmitter optical system of the telescope assembly. The intersecting point of the vertical and horizontal reticle lines of the collimator shall not exceed the tolerance specified in 3.8.1, as measured on the target. Record the collimation measurement.

4.6.7.2 Collimation - sighting optical system.- Conduct this adjustment in sequence with the test in 4.6.7.1. View through the eyepiece of the telescope assembly. The line of sight through the center of the reticle pattern shall meet the requirements as specified in 3.9. Record the collimation position, as measured on the target image. Any difference between the recorded measurement above and the recorded measurement established in 4.6.7.1 is the amount of collimation error. Collimation error shall not exceed the tolerance specified in 3.9.

4.6.7.3 Boresight adjustment, azimuth.- Perform this test from the reticle position established in 4.6.7.2. Rotate the azimuth adjustment screws to its clockwise stop then to its counterclockwise stop. This screw rotation shall meet the requirements specified in 3.9, as measured on the target image.

4.6.7.4 Boresight adjustment, elevation.- Conduct this test utilizing the procedure specified in 4.6.7.3. The elevation adjustment screw shall meet the requirements specified in 3.9, as measured on the target.

NOTE: Upon completion of these tests (4.6.7.3 and 4.6.7.4) the reticle shall be adjusted to the established position in 4.6.7.2.

4.6.7.5 Reticle accuracy.- This test shall be conducted in accordance with MIL-O-13830. The total angular subtense of the reticle graduations shall be as indicated on the reticle drawing. The reticle accuracy shall meet the requirements specified in 3.11, as measured on the target.

4.6.7.6 Eyepiece focus.- The test shall be made with the aid of a calibrated dioptometer with a magnification of at least 3 power (3X) and a diopter range for measuring the accuracies of the diopter knob scale as specified in 3.12. Adjust the dioptometer eyepiece for best focus of the dioptometer reticle and set the dioptometer objective scale to zero. Set the diopter scale of the telescope under test to zero diopter. Position the dioptometer at the telescope eyepiece assembly. While viewing through the dioptometer, obtain best focus of the telescope assembly reticle by adjustment of the dioptometer objective scale. The telescope reticle shall be in best focus within the limit specified in 3.12, as measured on the dioptometer objective scale. Rotate the telescope diopter knob to its clockwise stop, note diopter setting on scale, then rotate the diopter movement to its counterclockwise stop, note diopter setting on diopter scale. The requirements specified in 3.12 shall be met.

4.6.7.7 Parallax (sighting system).- Perform this test after parallax adjustment has been made in conformance with 3.13.1. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

diopter knob of the telescope shall be set for best focus of the telescope reticle. Parallax requirements shall be met when the target projector collimator is adjusted to the range specified in 3.13.1. Parallax will be recognized by any apparent displacement of the telescope reticle lines with respect to the target image lines when the observer's head is moved from side to side or up and down. Displacement between the telescope reticle and target image shall not exceed the tolerance specified in 3.13.1, as measured on the target.

4.6.7.8 Parallax (receiver system).- Conduct this test with the diaphragm (field-stop) illuminated as specified in 3.13.2. Remove the light source from the sighting system target collimator. Observe through the collimator eyepiece assembly. Parallax will be recognized by any displacement between the collimator reticle lines with respect to the diaphragm image while the observer's head is moved from side to side or up and down. The amount of parallax observed shall be within the tolerance specified in 3.13.2, as measured on the target collimator.

4.6.7.9 Parallax (transmitter system).- Perform this test with the target image illuminated with a filtered light source that conforms to 3.13.3. Observe through the Collimator I (Figure 1) as specified in 3.13.3. Parallax will be recognized by any displacement between the target image lines with respect to the collimator reticle lines when the observer's head is moved from side to side or up and down. Displacement between the collimator reticle lines and target image lines shall not exceed the tolerance specified in 3.13.3, as measured on the target.

NOTE: Where red light is specified in 3.13.3, a Corning No. 70 filter or equivalent may be utilized.

4.6.7.10 Focus (receiver system - field stop).- Perform this test with the field stop image illuminated as specified in 3.14. The test shall be made with the aid of a calibrated dioptometer with a magnification of at least 3 power (3X) and a diopter range for measuring the accuracy of the field stop focus as specified in 3.14. Adjust the dioptometer eyepiece for best focus of the dioptometer reticle and set the dioptometer objective scale to zero. Position the dioptometer at the telescope assembly as specified in 3.14. While viewing through the dioptometer, obtain best focus of the field stop image by adjustment of the dioptometer objective scale. The field stop image shall be in best focus within the tolerance specified in 3.14, as measured on the dioptometer objective scale.

4.6.7.11 Parallelism of reticle and image.- With the line of sight horizontal, position one extremity of the telescope vertical reticle line coincident to the vertical line of the target image. The telescope reticle line shall be parallel to the target image line within the tolerance specified in 3.17, as measured at the other extremity of the telescope vertical reticle line with respect to the vertical target image line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.6.7.12 Image tilt.- This test shall be performed with the aid of an auxiliary telescope of at least 3 power (3X) containing a vertical and horizontal reticle line. Plumb the vertical reticle line of the telescope and position the auxiliary telescope at the telescope eyepiece assembly. Observe through the auxiliary telescope and establish one extremity of the telescope vertical reticle line coincident to the vertical target image line. The other extremity of the telescope vertical reticle line at the vertical target image line shall be within the tolerance specified in 3.18, as measured on the target.

4.6.8 Diaphragm.- Perform this test with a "V" block capable of positioning the mounting bearings of the telescope assembly. The "V" block shall be secured to a leveled surface. Level the telescope utilizing a gage block and the locating pin used for alignment of the cavity assembly. A viewing collimator capable of measuring the field of view and reticle accuracies of the telescope, as specified in 3.19, shall be positioned and aligned to the geometric axis of the sighting telescope objective assembly. A light source shall be utilized to illuminate the diaphragm field of view aperture. Observe through the viewing collimator. Obtain coincidence of the collimator vertical and horizontal reticle lines with resepect to the vertical and horizontal telescope reticle lines. Any deviation between the diaphragm image with respect to the center square of the telescope reticle shall not exceed the tolerance specified in 3.19, as measured on the viewing collimator. Subsequent to the reticle alignment test, position the intersecting point of the vertical and horizontal reticle lines of the collimator to one extremity of the field of view. Record this measurement of the collimator reticle. Scan the field of view area to the other extremity. Record this position. The difference between the two recorded measurements shall not exceed the tolerance specified in 3.19, as measured with the viewing collimator.

4.6.9 Operability.- The tests in 4.6.9.1 through 4.6.9.3 inclusive shall be performed with standard torque measuring equipment provided with appropriate adapters to accommodate the assembly under test to determine conformance with 3.20.

4.6.9.1 Diopter adjustment torque.- With the adapter affixed to the diopter knob, rotate the knob through the range specified in 3.20.1. The running torque shall not be less than or exceed the values specified in 3.20.1, for the temperatures indicated in 3.4.2 and 3.4.3.

4.6.9.2 Potentiometer knob torque.- Conduct this test with the adapter affixed to the knob. The maximum torque required to rotate the knob shall not be less than or exceed the values specified in 3.20.2, for the temperatures indicated in 3.4.2 and 3.4.3.

4.6.9.3 Boresight adjustment torque.- Perform this test with the adapter affixed to the boresight adjustment screws. The maximum torque

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

required to rotate the azimuth and elevation screws shall not be less than or exceed the values specified in 3.20.3 for the temperatures indicated in 3.4.2 and 3.4.3.

NOTE: Upon completion of these tests, the reticle shall be positioned to the recorded measurement established in 4.6.7.2.

4.6.9.4 Parallax shield.- Operate the shield through its full range of travel. During this excursion, the shield shall conform to the requirements specified in 3.20.4, at the temperatures indicated in 3.4.2 and 3.4.3.

4.6.10 Reticle illumination.- Apply the operating voltage to the reticle lamp of the telescope as specified in 3.21. The illuminated reticle etchings shall be clearly defined when visually examined against a darkened background. Should the brightness of a reticle defect be greater than the reticle lines, the defect shall be cause for rejection. The potentiometer knob shall be rotated clockwise from the extreme counterclockwise position. During this rotation, the potentiometer shall meet the requirements specified in 3.21, while viewing the reticle pattern.

4.6.11 Interface requirements.

4.6.11.1 Housings.- Perform this test utilizing two test housings as specified in 3.22. One housing shall be designed to the maximum limit of each interface with a 10% of total limit, and one designed to the minimum limit of each interface with a tolerance of 10% of total limit. Do not extend any interface beyond maximum or minimum limits. All pertinent interfaces shall be provided as depicted on Drawings F10540705 and F10540967. The telescope assembly shall be capable of being properly positioned and secured to both test housings.

4.6.11.2 Support and caps.- Perform this test utilizing two test supports and caps as specified in 3.22. One support and caps shall be designed to the maximum limit of each interface with a 10% of total limit, and one designed to the minimum limit of each interface with a tolerance of 10% of total limit. Do not extend any interface beyond maximum or minimum limits. All pertinent interfaces shall be provided as depicted on Drawing F10540514. The telescope assembly shall be capable of being properly positioned and secured to both test support and caps.

4.6.11.3 Support.- Perform this test utilizing two test supports specified in 3.22, Drawing F11727551. One support shall be designed to the maximum limit of each interface with a tolerance of 10% of total limit and one designed to the minimum limit of each interface with a tolerance of 10% of total limit. Do not extend any interface beyond maximum or minimum limits. All pertinent interfaces shall be provided as depicted on Drawing F11727551. The telescope assembly shall be capable of being properly positioned and secured to both test supports.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.6.11.4 Connectors.- The telescope assembly shall be engaged to test connectors specified in 3.22 designed with pertinent interfaces conforming to requirements of drawings specified. Continuity between all connectors and test connectors shall be indicated when fully engaged.

5. PREPARATION FOR DELIVERY

(Not applicable)

6. NOTES

6.1 Intended use.- The telescope assembly is the central housing and support member of the Range Finder, Laser, AN/GVS-3 to which the left and right housing assemblies are secured. This housing contains the transmitting optics that direct the LASER beam to the target area and the sight-receiver optics that receive the target echoes and direct them to the electronic components of the right housing assembly and permit the operator to aim the range finder.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Order of document precedence
c. Provisions for First Article testing

Custodian:

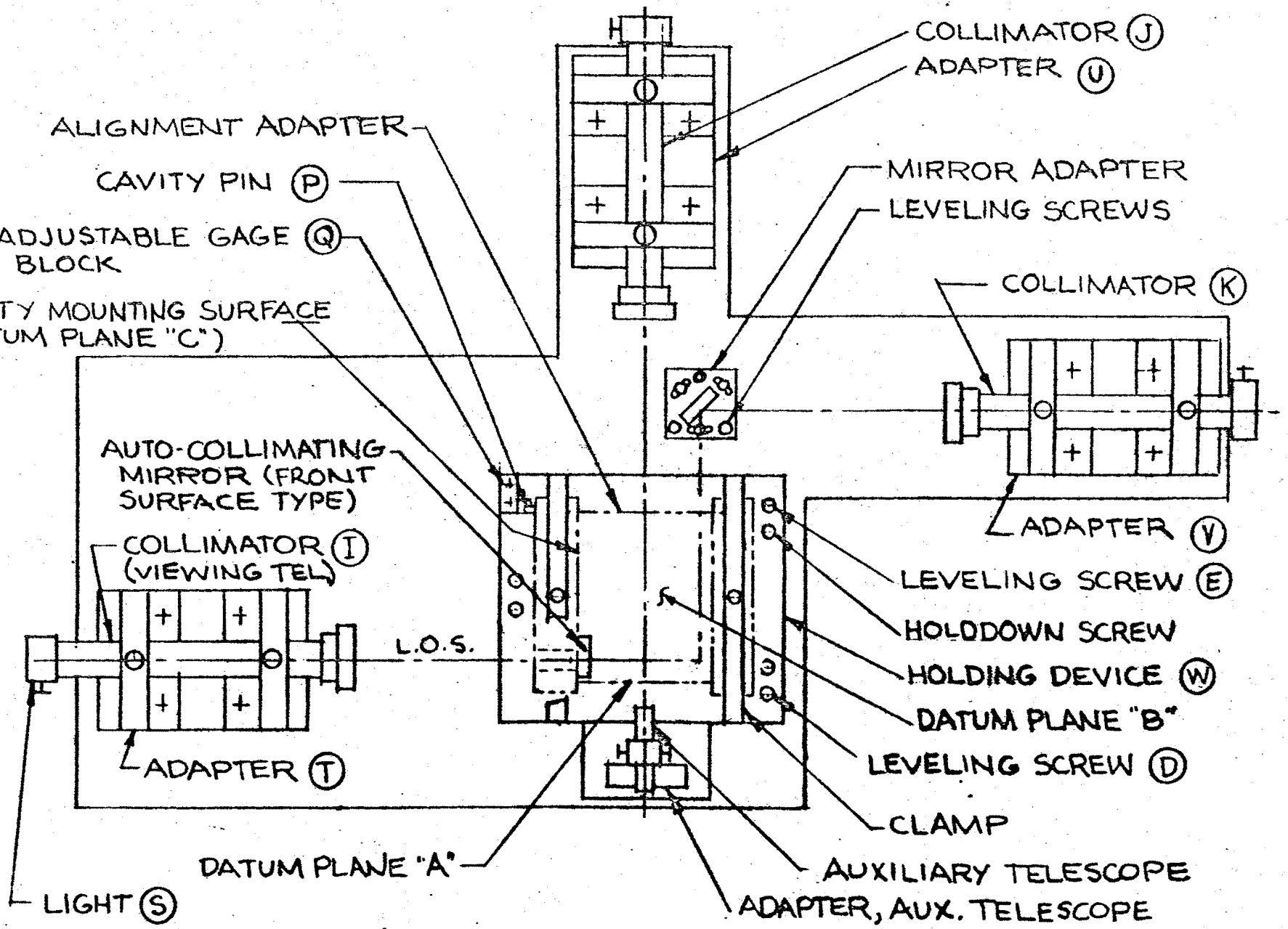
Army - MU

Preparing activity:

Army - MU

Project No. 1240-A219

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.



TELESCOPE ASS'Y TEST FIXTURE (RCV'R & XMTR)

FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.